Tutorial 7
Bending and Shear Stresses

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1. A solid steel rod of diameter d is supported as shown. Knowing that for steel $\gamma = 490$ lb/ft^3, determine the smallest diameter d that can be used if the normal stress due to bending is not to exceed 4 ksi.

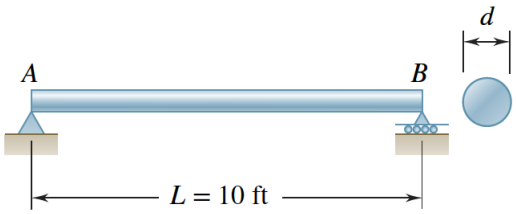


Fig. 1

2. The beam has a rectangular cross section and is subjected to a bending moment M. If the material from which it is made has a different modulus of elasticity for tension E_t, and for compression E_c as shown in Fig. 2, determine the location c of the neutral axis and the maximum compressive stress in the beam.

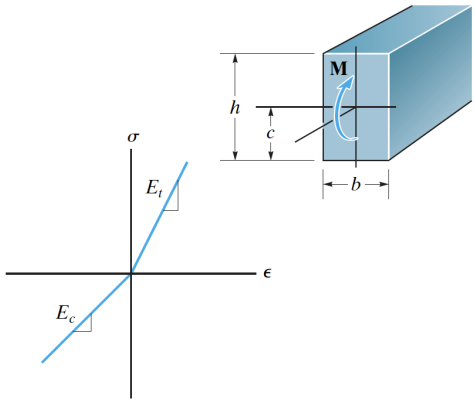


Fig. 2

3. The man has a mass of 78 kg and stands motionless at the end of the diving board. If the board has the cross section shown in Fig. 3, determine the maximum normal strain developed in the board. The modulus of elasticity for the material is $E = 125$ GPa. Assume A is a pin and B is a roller.

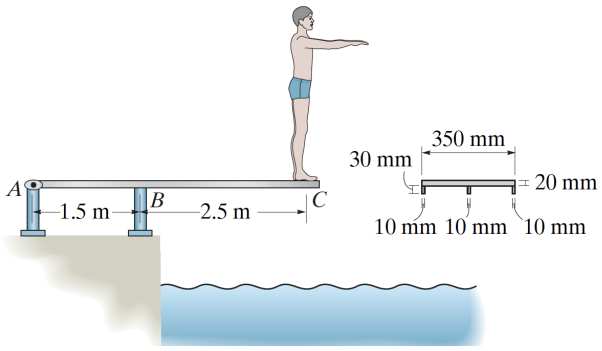


Fig. 3

4. Determine the moment M that should be applied to the beam shown in Fig. 4 in order to create a compressive stress at point D of $\sigma_D = 30$ MPa. Also sketch the stress distribution acting over the cross section and compute the maximum stress developed in the beam.

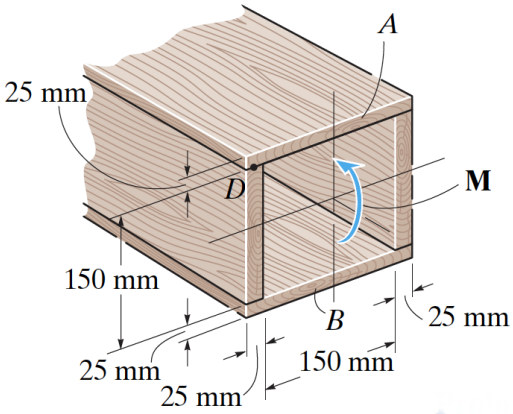


Fig. 4

5. A hollow box beam is constructed with webs of Douglas-fir plywood and flanges of pine, as shown in Fig. 5 in a cross-sectional view. The plywood is 1 in. thick and 12 in. wide; the flanges are 2 in. × 4 in. (nominal size). The modulus of elasticity for the plywood is 1,800,000 psi and for the pine is 1,400,000 psi.

   (a) If the allowable stresses are 2000 psi for the plywood and 1750 psi for the pine, find the allowable bending moment $M_{\max}$ when the beam is bent about the z axis.

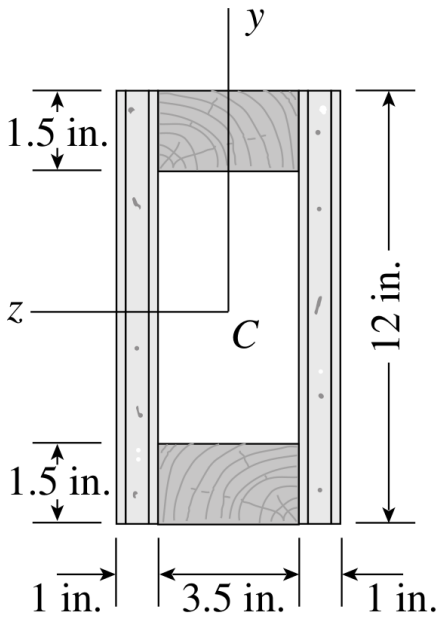


Fig. 5

6. A wooden cantilever can be assembled from 3 beams (dimensions of the cross section $b = a$ and $h = 2a$) in different ways. What is the maximal force F for the two variants 1 and 2 as shown in Fig. 6, if the maximal allowed shear stress in the bonding layer is given by τ_{allow}?

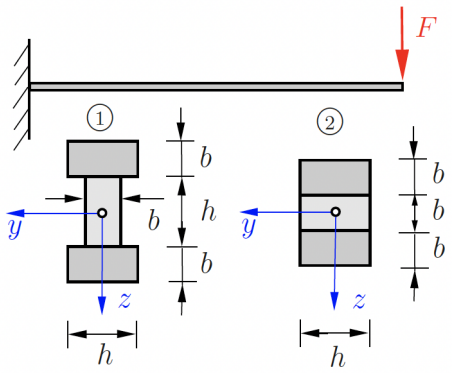
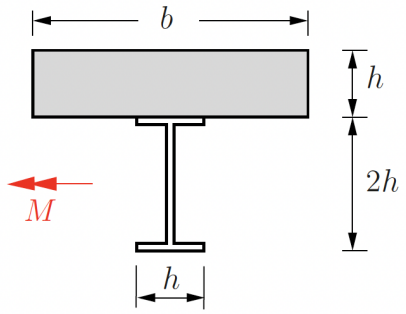


Fig. 6

7. A composite beam consists of an upper concrete slab and a steel I beam. The structure is loaded by a bending moment M as shown in Fig. 7.

   (a) Determine the width b of the concrete slab, such that compressive stresses occur only in the concrete part, while the tension is present in the steel part.

   (b) For this case compute the stresses in the extreme fibres of the two materials.



Given : $M = 1000$ kNm
$E_C = 3.5 \cdot 10^4$ N/mm^2
$E_S = 2.1 \cdot 10^5$ N/mm^2
$h = 40$ cm
$A_S = h^2 / 6$
$I_S = h^4 / 18$

Fig. 7

8. Knowing that for the beam shown in Fig. 8 the allowable stress is 12 ksi in tension and 16 ksi in compression, determine the largest couple M that can be applied?

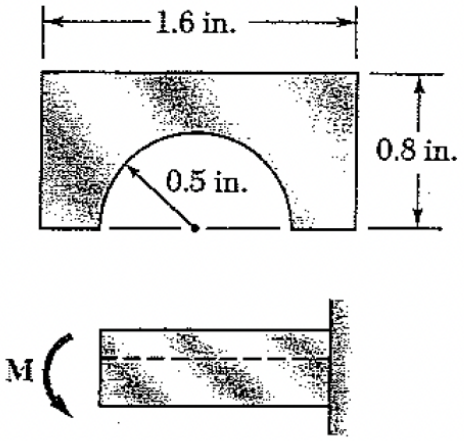


Fig. 8